THE ETALE TOPOS OF A SCHEME DOES NOT DEPEND ON THE CHOICE OF UNIVERSE FOR THE SMALL ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The same-chapter reference is located at its \label line inside the pinned chapter file; the three cross-chapter references are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Construction of the small etale site of a scheme from a big etale site. Same chapter, Definition environment labelled definition-tau-site (source line 1909, Stacks tag 03XB); the statement refers to it explicitly. Supplies the construction whose independence from the chosen universe is asserted; the comparison argument is the counted content.
- Reduction of a comparison of sites to a comparison of small categories and coverings. Topologies, Lemma topologies-lemma-contained-in (022J) and Lemma topologies-lemma-alternative (04HR), plus Sets, Lemma sets-lemma-what-is-in-it (000R); invoked at proof lines 2138, 2140 and 2149. Supply the reduction to the case of nested big sites, the smallness fact that every affine scheme etale over S sits in both, and the criterion that an equivalence preserving standard coverings induces an equivalence of topoi.

Source reference. Original definition 03XB: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1909–1909. Same-chapter pointer named in the counted statement (line 2131) for the construction whose independence is asserted; the comparison argument is retained verbatim in the counted span.

Source reference. Original lemma 022J: sources/stacks-chow/tags.txt, source lines 2708–2708. Cross-chapter standard prerequisite invoked once at proof line 2138 to reduce to the case of nested big etale sites.

Source reference. Original lemma 000R: sources/stacks-chow/tags.txt, source lines 36–36. Cross-chapter standard prerequisite invoked once at proof line 2140 for the smallness statement that an affine scheme etale over S belongs to both big sites.

Source reference. Original lemma 04HR: sources/stacks-chow/tags.txt, source lines 5877–5877. Cross-chapter standard prerequisite invoked once at proof line 2149 for the criterion that an equivalence of small categories preserving standard coverings

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induces an equivalence of topoi; the verification of that preservation is retained in the counted span.

Lemma 0.1. *Let S be a scheme. The étale topos of S is independent (up to canonical equivalence) of the construction of the small étale site in Definition 03XB.*

Proof. We have to show, given two big étale sites $Sch_{étale}$ and $Sch'_{étale}$ containing S, then $Sh(S_{étale}) \cong Sh(S'_{étale})$ with obvious notation. By Topologies, Lemma 022J we may assume $Sch_{étale} \subset Sch'_{étale}$. By Sets, Lemma 000R any affine scheme étale over S is isomorphic to an object of both $Sch_{étale}$ and $Sch'_{étale}$. Thus the induced functor $S_{affine,étale} \to S'_{affine,étale}$ is an equivalence. Moreover, it is clear that both this functor and a quasi-inverse map transform standard étale coverings into standard étale coverings. Hence the result follows from Topologies, Lemma 04HR. □